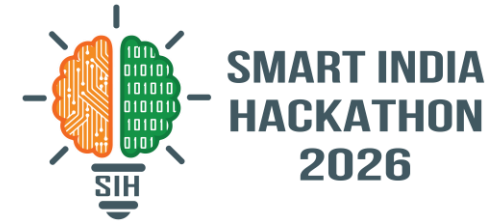
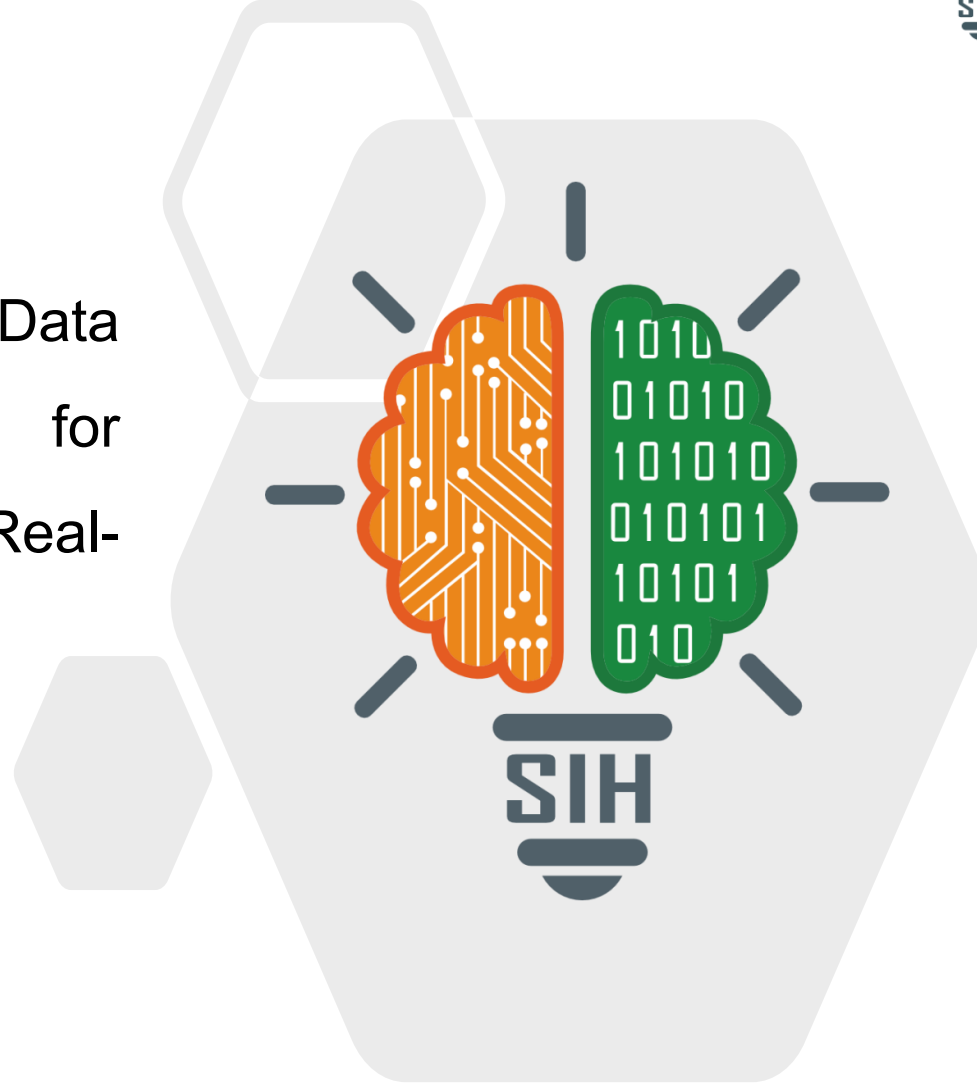


SMART INDIA HACKATHON 2026



- **Problem Statement ID** – SIH26122
- **Problem Statement Title**- Intelligent Data Capture & Schedule-Linking Layer for Infrastructure Project Management: Real-Time Actual Progress Tracking
- **Theme**- Smart Automation
- **PS Category**- Software
- **Team Name**- Innovators United



Problem Summary

- In EPC projects, the **disconnect** between **unstructured daily field reports** and **rigid EPC schedule IDs** creates a **manual reconciliation bottleneck** that **delays progress tracking** by weeks and **cripples real-time risk analytics**.

Our Proposed Solution: SynchroLink Layer

- Multimodal Ingestion Agent:** A low-friction interface (voice, text, or file upload) where supervisors log daily progress in their natural language, which an AI instantly parses into structured data.
- Hierarchical AI Matcher:** An intelligent routing engine that takes the parsed field report and uses semantic search to find the exact matching task ID in the massive master schedule.
- Confidence-Gated Sync:** High-confidence matches automatically update the official schedule's Gantt charts in real-time.
- Planner Triage Hub:** Low-confidence matches are flagged in a dashboard for human review, ensuring no data is hallucinated or lost.

Unique Value Proposition



Context-Aware Bounding: We filter by engineering discipline and plant location before matching, drastically reducing AI hallucinations.



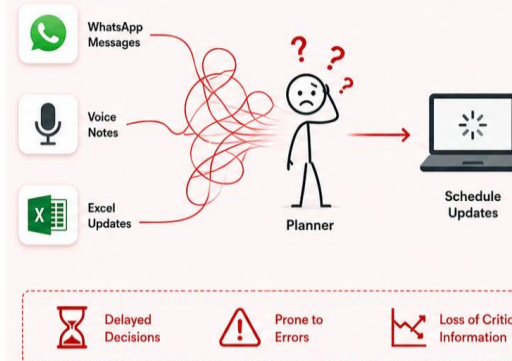
Human-in-the-Loop Architecture: Safe automation. We don't blindly overwrite project databases; ambiguous data requires 1-click human validation.



Delay Analytics Database: We capture root causes of delays, building a queryable historical database to help predict risks on future projects.

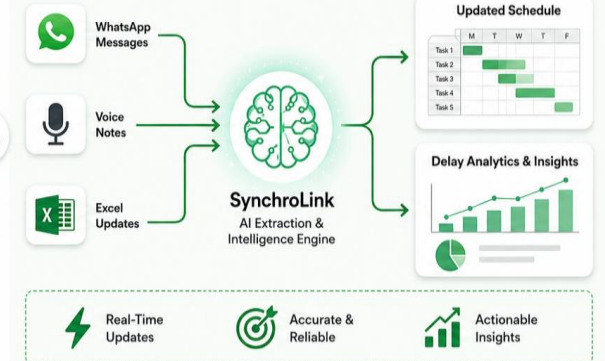


Current Reality: Manual, Delayed, Lossy.



VS.

Our Solution: Automated, Real-Time, Data-Driven.

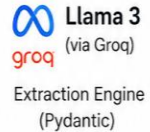


Tech Stack

Core Backend and API Layer



AI & NLP



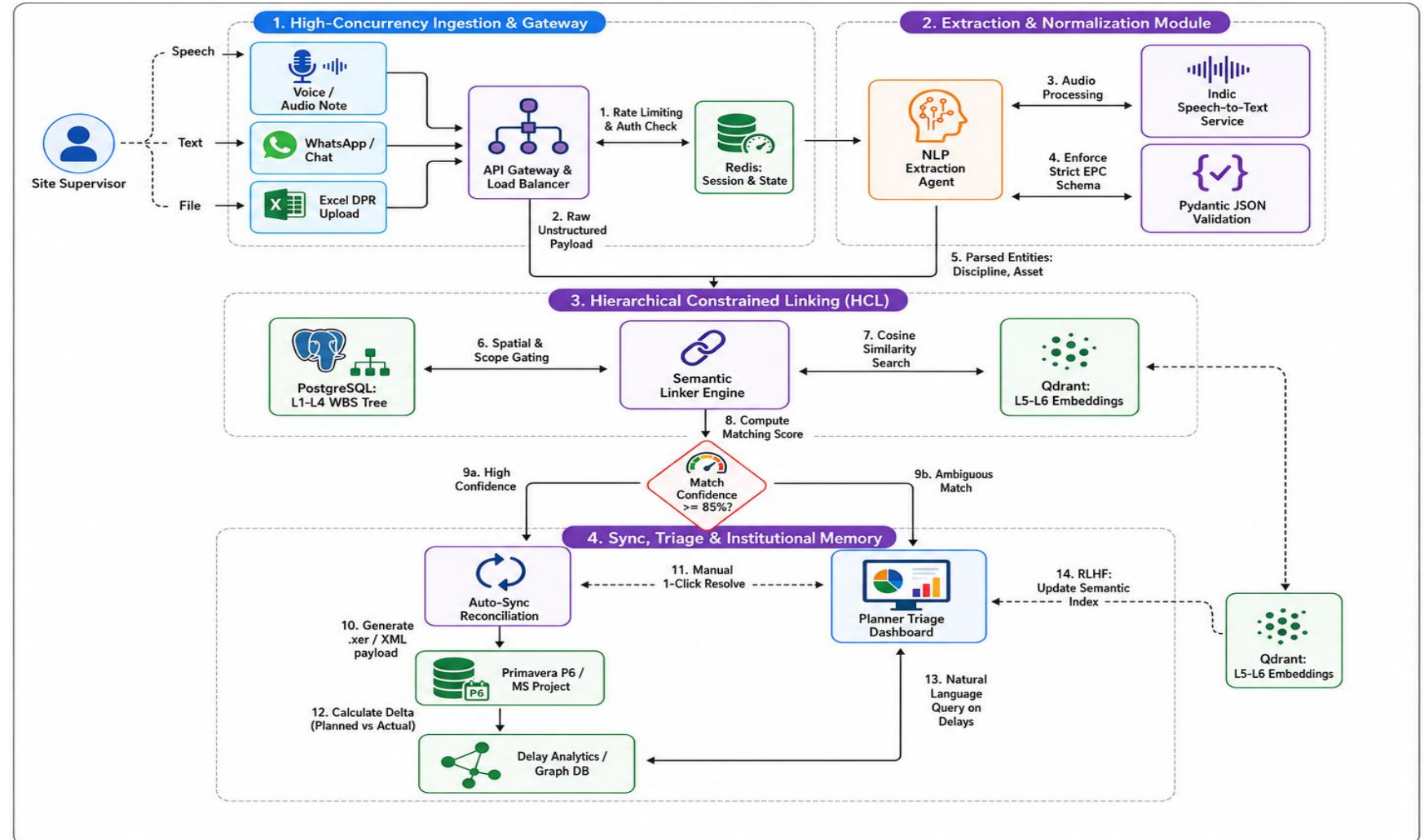
Databases & Infrastructures



Frontend Interface



Architecture Diagram



Technical Feasibility

High-Performance Backend: Rust-based distributed API Gateway to handle high-concurrency batch uploads (Excel/CSV DPRs) without bottlenecking, coupled with a Python/FastAPI microservice for LLM orchestration.

Deterministic AI Extraction: Open-source LLMs combined with strict Pydantic schemas enforce rigid JSON output for EPC variables (Discipline, Asset ID, Action), completely eliminating free-text agentic latency.

Hybrid Database Architecture: Proven PostgreSQL implementation for querying the rigid, hierarchical WBS parent-child tree, paired with a Vector DB (Qdrant/FAISS) for lightning-fast semantic similarity searches.

Standardized PMIS Integration: Direct generation of standard .xer and MS Project XML payloads ensures seamless bi-directional syncing with enterprise Primavera P6 instances.

AI Hallucinations

Vernacular & Messy Field Inputs

Destructive Schedule Overwrites

Challenges & Mitigations

Hierarchical Constrained Linking: Filters by Discipline and Plant Unit *before* vector search, **strictly bounding the AI's search space** to prevent false positives.

Integration of **robust speech-to-text pipelines** (e.g., AI4Bharat/IndicWhisper) optimized for code-mixed audio to **capture local site jargon** accurately.

Human-in-the-Loop Architecture: A strict **confidence threshold ($\geq 85\%$)**. Anything **lower bypasses auto-sync** and is safely **routed to the Planner Triage UI** for 1-click manual verification.

Scalability & Expansion

Enterprise Extensibility



Seamlessly expandable beyond Primavera P6 to integrate with standard SAP PMIS and Oracle ERP modules via RESTful adapters.

Cross-PSU Deployment



The architecture is domain-agnostic enough to scale across other heavy-infrastructure entities (ONGC, GAIL, L&T).

Predictive Risk Modelling



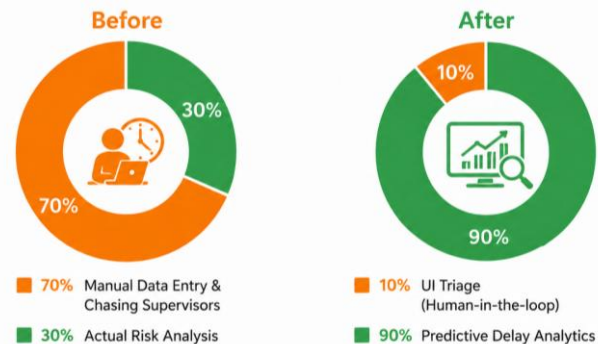
The "Institutional Memory" database tracking history lays the foundation for training predictive AI models to forecast downstream project delays.

Schedule Update Latency



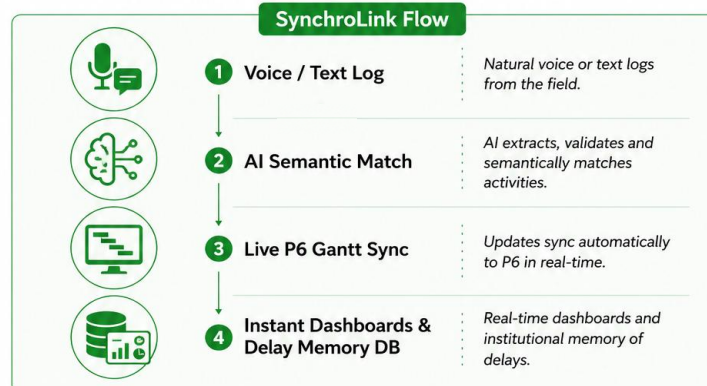
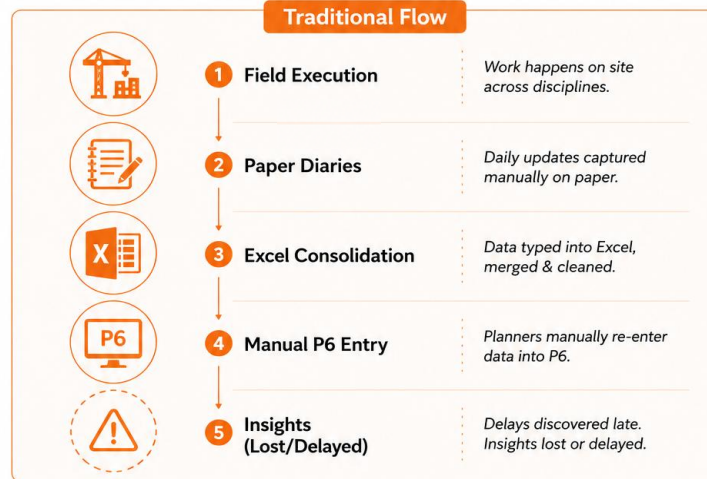
 Accelerates project tracking by **95%**, turning weeks of manual data entry into instant API-driven updates.

Project Planner Bandwidth



 Frees up highly-paid planning engineers to focus on **proactive risk mitigation** instead of clerical data matching.

The Process Transformation



 Transforms a slow, error-prone process into an intelligent, real-time data backbone for **proactive project control**.

Economic & Contractual Impact

Resolves Contractor Disputes: Immutable, timestamped actuals prevent disputes over RA (Running Account) Bills and Extension of Time (EOT) claims.

Prevents Cascading Penalties: Identifies micro-delays in real-time so managers can reallocate resources and avoid multi-million-rupee LD penalties.

Institutional Memory Capture: Boosts root-cause capture from <15% to 90%+, building a historical database for accurate PSU project pricing and planning.

Strategic Alignment

Direct Alignment with MoPNG Goals: Democratizes project tracking for non-technical field workers (using vernacular/Hinglish voice notes).

Zero-Friction Adoption: Requires zero training for ground staff. They report exactly as they do now, while the AI handles the enterprise mapping.

EPC Domain & Project Controls Research

- **Standardizing WBS Hierarchies:**
 - *Oracle Primavera P6 EPPM Documentation:* Referenced for standardizing .xer and XML schema parsing for Level 5 and Level 6 Work Breakdown Structure (WBS) activity relationships.
 - *Project Management Institute (PMI):* Guidelines on Schedule Variance (SV) and delay root-cause analysis in heavy infrastructure.
- **Construction NLP Literature:**
 - *Reference Study:* "Automated Information Extraction from Construction Daily Reports using NLP" (Journal of Computing in Civil Engineering). Validated our approach to extracting *Discipline*, *Action*, and *Asset Tags* from unstructured field diaries.

AI Models & Semantic Architecture

- **Hybrid Semantic Search & Entity Linking:**
 - *Pinecone/Qdrant Documentation:* Research on combining BM25 (Lexical) and Dense Cosine Similarity (Vector) for high-precision entity mapping to prevent AI hallucinations.
- **Indic Vernacular Speech-to-Text:**
 - *AI4Bharat (IIT Madras):* Utilized research on IndicWhisper for transcribing code-mixed regional audio (Hinglish/Assamese) from site supervisors.
- **Deterministic LLM Extraction:**
 - *Pydantic & Constrained Decoding:* Applied schema-enforced generation methodologies to ensure the LLM output strictly adheres to the EPC database schema without generating arbitrary text.

Core Tech Stack References

- **High-Performance Backend:** Rust (tokio ecosystem) for the API gateway handling concurrent daily progress report (DPR) file ingestion.
- **Microservices API:** Python (FastAPI) asynchronous documentation for orchestrating the LLM and Vector DB pipelines.
- **Frontend Data Visualization:** React & Gantt charting libraries for the Planner Triage UI.